

Best-in-class integration technologies: this ERP × a HIS (TrakCare)

A companion to the demo speaker notes — the standards, architecture pattern, and phased path for connecting this ERP's Party / Product / Order / Facility / Accounting core to InterSystems TrakCare, without either system becoming a satellite of the other.

IN ONE PARAGRAPH

The standards choice isn't really open anymore: **HL7 FHIR R4** is the modern default, and TrakCare's own InterSystems IRIS platform already speaks it natively — plus legacy HL7 v2.x, which still carries most live ADT traffic in production HIS deployments. **NPHIES is not our problem to solve at all:** InterSystems is independently certified NPHIES-compliant and TrakCare already submits claims through it for live Saudi hospitals today. That removes an entire integration leg from this design. What's actually left to build is small: a thin FHIR-shaped adapter added to this ERP's existing API project, talking directly to IRIS's own interoperability engine — which already does the queuing, retries, and routing our side would otherwise have had to build from scratch.

01 GLOSSARY — IF HIS TERMS ARE NEW TO YOU

Every term used on this page, in plain language, grouped by what it's actually for. Skip this section entirely if you already work in health IT.

THE SYSTEMS THEMSELVES

HIS Hospital Information System

The hospital's own software — admissions, orders, results, and clinical billing. TrakCare is one specific HIS product, made by a company called InterSystems.

EMR / EHR Electronic Medical / Health Record

The part of a HIS that holds a patient's actual chart — diagnoses, notes, medications. People often say "EMR" and "HIS" interchangeably in conversation.

ERP for contrast

This system. Manages the business side — orders, inventory, accounting — rather than clinical care. The two systems' data has to meet in the middle.

PACS / RIS

The systems that store medical images (X-rays, scans) and run the radiology department's workflow. Mentioned only so it's clear this ERP never touches image data directly.

HOW SYSTEMS TALK TO EACH OTHER

HL7 Health Level Seven

The umbrella name for the family of hospital messaging standards. Not one format — a whole lineage of them, including the next two entries.

HL7 v2.x

The older, still-dominant message format — plain text with fields separated by | pipe characters. Not pretty, but it's what most live hospital systems actually send today.

ADT Admit / Discharge / Transfer

An HL7 v2 message announcing patient movement — admitted, discharged, or moved to a different ward or bed.

ORU / ORM Observation Result / Order

Two more HL7 v2 message types — ORU carries a lab or test result, ORM carries a new order such as "run this test."

FHIR

Fast Healthcare Interoperability Resources, said "fire"
The modern replacement for HL7 v2 — structured JSON exchanged over a normal web API, the same style this ERP's own OData endpoints already use.

Resource a FHIR term

A named, structured record type — Patient, Encounter, Claim. Think of it as a DTO with an industry-agreed shape everyone builds to.

MATCHING UP IDENTITY ACROSS SYSTEMS

MRN Medical Record Number

The HIS's own unique ID for a patient — the hospital-side equivalent of this ERP's PartyId.

IHE Integrating the Healthcare Enterprise

An initiative that defines standard patterns for using HL7/FHIR consistently, so two vendors' systems can actually interoperate, not just both claim "FHIR support."

PIX / PDQ two IHE patterns

PIX keeps two systems' patient IDs cross-referenced to each other; PDQ lets one system ask the other "does this patient already exist, and under what ID?"

SAUDI-SPECIFIC & REGULATORY

NPHIES

Saudi Arabia's national platform that every hospital and insurer must connect through for eligibility checks, approvals, and claims. TrakCare already has this certified — it's TrakCare's job, not this ERP's.

PDPL Personal Data Protection Law

Saudi Arabia's data-privacy law — governs how personal and health data must be stored, moved, and protected.

Payer

Insurance terminology for whoever is paying the bill — an insurance company, in most cases here.

Eligibility / Prior auth

Asking the payer "is this patient covered, and will you approve this treatment" — before or during care, via NPHIES.

Claim

The bill submitted to the payer after treatment is given.

Remittance

The payer's response to a claim — what they agreed to pay, and why anything was reduced or rejected.

SECURITY

OAuth2

A standard way for one system to prove to another "I'm allowed to access this," using short-lived tokens instead of a shared password.

SMART on FHIR

OAuth2 adapted specifically for healthcare FHIR APIs — the standard way a clinical app authenticates itself.

Mutual TLS mTLS

Both sides of a connection prove their identity with certificates — not just the client checking the server, the way an ordinary "padlock icon" website connection works.

TRAKCARE / INTERSYSTEMS SPECIFIC

InterSystems IRIS

The underlying database-and-integration platform TrakCare is built on — not just a hospital app, but also the "engine room" capable of sending and receiving HL7/FHIR messages.

Interoperability engine formerly "Ensemble"

The piece of IRIS that translates and routes messages between TrakCare and outside systems, with queuing and retry already built in — the thing our FHIR adapter in §03 talks to directly, no extra infrastructure needed on our side.

CLINICAL/BUSINESS TERMS USED IN §04'S MAPPING TABLE

Encounter

One clinical visit or episode — a checkup, an ER visit, a hospital stay. Roughly the HIS equivalent of an ERP order.

Coverage

A patient's insurance policy details, in FHIR form.

Practitioner / RelatedPerson

FHIR's terms for a clinician, and for a patient's family member or guardian, respectively.

ServiceRequest

A FHIR resource for an ordered procedure or test — the FHIR-shaped version of an ORM message.

ChargeItemDefinition

A FHIR resource describing something billable and its price — the FHIR-shaped version of a product/price-list entry.

ExplanationOfBenefit EOB

A FHIR resource showing what the payer decided about a claim — the FHIR-shaped version of "remittance."

02 STANDARDS LANDSCAPE

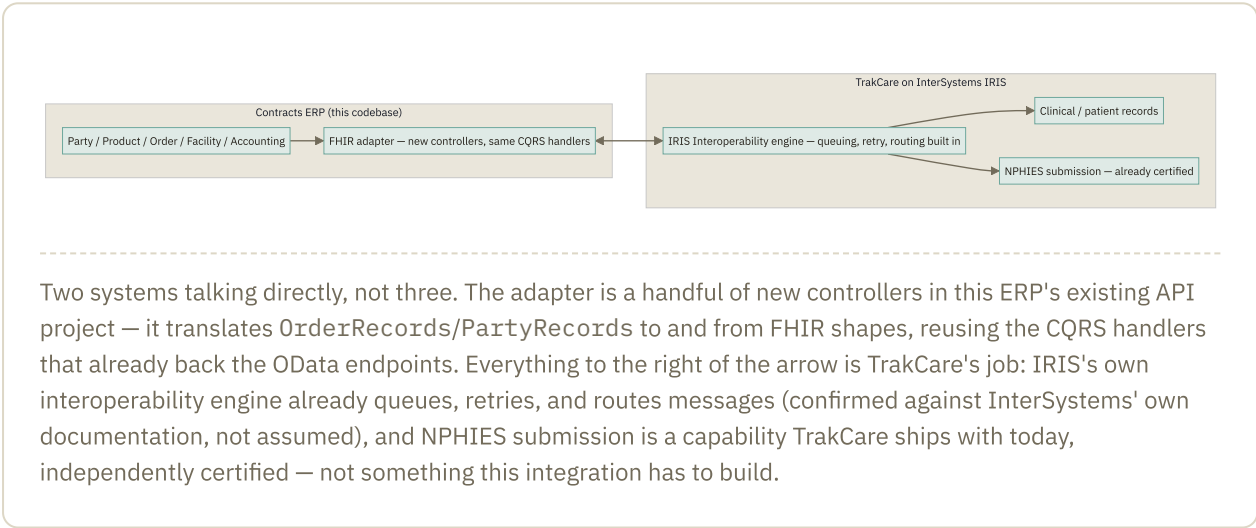
What each one actually carries, and whether it's live, optional, or out of scope for an ERP-side integration.

STANDARD	CARRIES	STATUS HERE	WHY IT MATTERS
HL7 v2.x ADT / ORU / ORM	Real-time clinical events — admit / discharge / transfer, orders, results	live in prod	Still the actual wire format for most ADT feeds today, TrakCare included — a FHIR-first design still has to speak it at the edge
HL7 FHIR R4	REST/JSON resources — Patient, Encounter, Coverage, Claim, Invoice	TARGET STANDARD	What any new integration should be built on; IRIS Interoperability exposes it natively
NPHIES	Saudi national exchange — eligibility, prior-auth, claims, remittance	TRAKCARE'S JOB, NOT OURS	InterSystems is independently NPHIES-certified and TrakCare already submits claims through it for live Saudi hospitals — our adapter never has to speak NPHIES directly
IHE PIX / PDQ	Cross-system patient identity	recommended	Keeps PartyId and TrakCare's MRN from drifting apart once both

STANDARD	CARRIES	STATUS HERE	WHY IT MATTERS
	resolution & query		systems can write identity
DICOM	Imaging	out of scope	A HIS/RIS/PACS concern — the ERP side of this partnership never touches image data directly

03 RECOMMENDED ARCHITECTURE — SIMPLER THAN THE FIRST DRAFT

No new service to deploy, no event bus to run and monitor. One new set of endpoints inside this ERP's existing API project, talking straight to the integration engine TrakCare already operates.



04 DOMAIN DEEP-DIVE — THE SAME TABLES, A DIFFERENT VERTICAL

This is the part worth saying out loud in the room: nothing below is a new table or a new field. Each domain already holds a property-development vertical (and, in Product's case, a manufacturing one too) — the columns below show it could hold TrakCare's world just as easily, sitting in the exact same schema.

Party **PartyRoleTypeId** FHIR: **Patient** **RelatedPerson** **Practitioner**

One Party entity, Person or PartyGroup — every role is just a PartyRole row against an open lookup, not a different table.

ALREADY LIVE HERE

Property buyer Sales broker Employee
Building contractor Material vendor

SAME TABLE, TRAKCARE'S WORLD

Patient Referring physician
Attending consultant Ward nurse
Lab technician Next of kin / legal guardian
Payer — Bupa Arabia, Tawuniya, MedGulf
MOH / regulator

ALREADY IN THE SCHEMA **PartyQual** + **PartySkill** already carry a qualification type with its own **VerifStatusId**, and years-of-experience plus a proficiency rating — ready for clinician credentialing (board certification, verification status, specialty registry) with zero schema change. **PartyIdentification**'s open type+value pair covers a national ID, an Iqama, or a payer member ID the same way a new lookup row does today. **PartyBenefit** already models a two-party, two-role benefit record with cost and period — a payer benefit plan between patient and insurer, out of the box.

Product **ProductTypeId** FHIR: **ChargeItemDefinition**

One Product table gated by **ProductTypeId** — the same row shape that already holds an apartment and a bag of cement holds a drug, an implant, or a bed-day.

ALREADY LIVE HERE

Apartment unit Cement Steel rebar
Finished villa Manufactured component

SAME TABLE, TRAKCARE'S WORLD

Paracetamol 500mg tablet
Amoxicillin 250mg/5ml suspension
Hip implant — titanium
MRI — brain with contrast
CBC (complete blood count) ICU bed-day
General ward bed-day
Nursing consultation — 30 min
N95 mask, box of 50

ALREADY IN THE SCHEMA **ProductCategoryMember** is many-to-many already — the same apartment can sit in a 2nd category live on screen, which for TrakCare means a formulary tier and a drug class at once. **ProductPrice** already supports concurrent, dated, per-currency prices — payer-contracted rates by agreement and date. **ProductAssoc** (MANUF_COMPONENT) already expresses a bill-of-materials — a surgical procedure linked to its required supply kit. **ProductFeature** already picks size/color/spec variants — dosage, formulation, route of administration.

Order / Encounter OrderTypeId FHIR: Encounter ServiceRequest

One `OrderHeader/OrderItem` engine, driven by `OrderTypeId` and a status lifecycle — sales, purchase, and quote are already the same machine, so a clinical encounter is too.

ALREADY LIVE HERE

Apartment sales order

Cement purchase order

Villa sales quote

Manufacturing sales order

SAME TABLE, TRAKCARE'S WORLD

Inpatient admission — 3-day stay, Cardiology

ER visit

Outpatient clinic visit

Lab order — CBC + lipid panel

Radiology order — chest X-ray

Surgical case — appendectomy, OR-3

Home-health visit

Telehealth consultation

ALREADY IN THE SCHEMA The same status lifecycle that already tracks an order from created to approved to fulfilled already models a clinical order's real lifecycle — ordered, in-progress, resulted — without inventing a single new state.

Facility FacilityTypeId FHIR: Location

Facility models warehouse, store, and plant generically; `FacilityLocation` gives it an internal hierarchy — aisle/shelf/bin today, ward/room/bed tomorrow.

ALREADY LIVE HERE

Sales office

Materials warehouse

Cement plant

Show unit / model apartment

SAME TABLE, TRAKCARE'S WORLD

Cardiology ward — Ward B, 3rd floor

ICU

OR-3 operating theatre

Bed 204-B

Radiology department

Central pharmacy

ER triage bay

Blood bank

ALREADY IN THE SCHEMA A ward, a room, and a bed are a new **FacilityType** plus a naming convention on the existing hierarchy — not a new concept, and not a new table.

WorkEffort WorkEffortTypeId

OFBiz's generic "anything that happens over time" — the same entity already tracks a construction project and a manufacturing run.

ALREADY LIVE HERE

Apartment construction project

Manufacturing production run

SAME TABLE, TRAKCARE'S WORLD

Post-op recovery pathway

Surgical case workflow

Nursing shift / rounds

Infection-control isolation period

Care coordination episode

GL / Accounting

AcctgTransTypeId

FHIR:

Claim

ExplanationOfBenefit

One accounting-transaction engine posts everything into one chart of accounts today, regardless of source module — a claim posts through the identical engine an apartment installment does.

ALREADY LIVE HERE

Apartment installment payment

Project revenue/cost recognition

Broker commission payable

SAME TABLE, TRAKCARE'S WORLD

Insurance claim — inpatient stay

Co-payment collected at discharge

Consumables cost allocated to a department

Physician fee-split

Refund for a rejected claim line

ALREADY IN THE SCHEMA Budget/BudgetItem — a full header-plus-line-item schema with review, revision, and approval sub-entities alongside — has exactly one reference in the whole codebase today. Schema complete, ready for departmental or payer-contract budgets, nothing built on it yet.

Flow direction, in short: Party is bidirectional (ERP owns the billing party, TrakCare owns clinical identity); Product, Order, and Facility flow TrakCare → ERP (their catalog and encounters become our pricing and orders); GL flows ERP → TrakCare → NPHIES → payer, with remittance posting back the same path into the GL engine already in place.

05 WHY THIS FITS THE CURRENT STACK

No new service to stand up, patch, or monitor — the adapter is new endpoints in the existing API project, thin because the ERP is already shaped like a REST API, not a database dump.

OData endpoints (OrderRecords, PartyRecords, QuoteRecords) are already resource-shaped, filterable lists — the adapter translates them, it doesn't replace them.

CQRS already separates reads from writes. FHIR's own verbs map directly: GET → Query, POST/PUT → Command — no new mental model for the handler layer.

The MediatR pipeline already runs a LoggingBehavior ahead of every handler — that's the natural slot for FHIR resource validation too, without touching a single existing handler.

06 SECURITY & COMPLIANCE

OAuth2 / SMART on FHIR

Token-scoped access between systems — no shared database credentials, no static API keys

Mutual TLS

Between the ERP's adapter and IRIS directly — one hop, since there's no separate facade tier in between

PDPL alignment

Saudi Personal Data Protection Law — data residency and consent, layered onto the audit trail this ERP already writes to `AUDIT_ACTIVITY`

Field-level encryption

National ID and clinical identifiers encrypted in transit and at rest, not just the connection

07 PHASED ROADMAP

Identity first, money last — each phase is independently useful and none requires the next to already exist.

Phase 1

Identity sync

Party ↔ Patient via HL7v2 ADT or FHIR Patient, one direction: TrakCare → ERP

Phase 2

Encounter / order sync

Billable encounters land as ERP sales orders automatically, same OrderHeader/Item engine

Phase 3

Billing handoff

ERP invoices reach TrakCare in FHIR Claim shape; TrakCare submits to NPHIES itself and posts remittance back to GL

Phase 4

Push-based sync

IRIS pushes to the adapter as events happen, replacing scheduled polling — no new infrastructure, just a subscription IRIS's engine already supports

WHERE THIS COMES FROM

§04's domain deep-dive expands on the live-demo walkthrough in the companion speaker notes ("One Core, Two Verticals") with the fuller set of dormant-capability proof points and concrete TrakCare-world examples for the room. The simplification in §03 is verified, not assumed: InterSystems' own documentation confirms IRIS's interoperability productions already do message queuing, priority routing, and configurable retry — and InterSystems has been independently certified NPHIES-compliant since 2022, with TrakCare live on it

GUARDRAIL — DON'T OVERCLAIM

None of §03–07 is built today. This is the recommended target architecture, not a status report. If pressed for a committed date, land on the phased roadmap in §07 rather than a single number.

at real Saudi hospitals. Also grounded in the deeper KSA ERP×HIS integration design this team reviewed in July 2026 — 204 fully-specified integration points, 816 field-level mappings, dedicated ZATCA and NPHIES chapters.

